

Aadi Desai

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Education

Imperial College London

Oct 2019 – Jun 2023

Master of Engineering in Electronics and Computer Engineering | **First Class with Honours**

Skills

Rust, Go, Git, Linux, Cloudflare Workers, TypeScript, Docker, Python, C++, SQL, SystemVerilog, VHDL, Solidity, F#

Projects

IRC-style Chat Server and Terminal Client GitHub: [supleed2/go-chat](https://github.com/supleed2/go-chat)

Dec 2023 – Jan 2024

- Built a chat server and client, with an interactive terminal user interface following the Elm (MVU) architecture
- Shared types imported to client and server code, maintaining a single source-of-truth for communication
- Server maintains chat history for multiple rooms, clients connect via WebSockets and exchange JSON messages
- Go learning project, including WebSockets, Goroutines and Channels for message passing
- Used mutexes to avoid race-conditions and concurrency bugs when shared data is mutable

Automated Verification Discord Bot GitHub: [supleed2/nanobot](https://github.com/supleed2/nanobot)

Jul 2023 – Aug 2023

- Developed a Discord bot for member onboarding, built on the [Poise](#) framework and the Shuttle platform
- Replaces manual checks for Imperial student status, as busy periods resulted in many missed join requests
- Running since August 2023 with minimal downtime, allowing members to join quickly and independently
- Leverages Cloudflare Pages + Functions to query the university API without storing sensitive login credentials
- Rust learning project in a real-world environment, with active users providing feedback and feature requests

Final Year Project: FPGA Accelerator for StackSynth GitHub: [supleed2/EIE4-FYP](https://github.com/supleed2/EIE4-FYP)

Oct 2022 – Jun 2023

- [LiteX](#) + SystemVerilog project to generate audio, extending the Imperial 3rd Year EE Embedded Systems module
- LiteX wrapper module connecting the custom designs and existing IP, e.g. the integrated RISC-V CPU, VexRiscV
- Designed SystemVerilog modules which are instantiated in the LiteX wrapper: a CAN-bus receiver to accept messages from connected StackSynth boards, a 64-channel 48kHz configurable wave sample generator

Experience

Software Engineer – Ultra Intelligence & Communications

Sep 2024 – Present

- Built an async Rust watchdog service, diagnosed and resolved cancel-safety issues of Axum request handling
- Extended strategic proof-of-concept prototypes to stress-test and demonstrate early-stage projects, transforming ideas into working products to validate potential product opportunities
- Implemented Quantum-resistant cryptographic primitives, with side-channel and memory safety assurances
- Created a bus-based FPGA debug port, providing read/write access to user-defined registers within VHDL designs, for automated testing and programmatic access to signals in real-time

Junior Embedded Software Engineer – Prolec Ltd

Apr 2024 – Aug 2024

- Optimised automated safety systems for construction and excavation equipment, including the ability to restrict power and prevent equipment from entering dangerous areas or reaching unstable positions
- Overhauled the C++ codebase to reduce code size, via conditional compilation and reduced class inheritance, providing space for new features on existing space-constrained hardware
- Implemented the J1939 messaging protocol atop the CAN bus, including handshakes to confirm safety-critical requests, allowing the product to run headless with an external display

Digital Design Engineer – Nordic Semiconductor

Apr 2022 – Sep 2022

- Developed open-source tooling for syntax highlighting and configurable style / syntax linting of SystemVerilog
- Updated SystemVerilog simulation designs to use only synthesizable constructs, for Verilator compatibility
- Gathered feedback from colleagues for improvements to the editor tooling, including linting rules to add